

CSA 1718 – Artificial Intelligence

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3. Prolog Program to Suggest Diet Based on Disease

Aim

To write and implement a Prolog program to suggest a suitable diet based on a given disease.

Objective

- To understand rule-based decision systems.
- To associate diseases with suitable diet suggestions.
- To retrieve the suggested diet using a query.

Theory

A rule-based system uses predefined facts and relationships to produce a recommendation. In this program, each disease is associated with a general diet category.

This is only a **simple educational rule-based example**, not medical advice.

Algorithm

1. Define diseases as facts.
2. Associate each disease with a diet.
3. Define the suggest_diet rule.
4. Give a disease as input.
5. Prolog returns the corresponding diet.

Program

```
diet(diabetes, low_sugar).
```

```
diet(obesity, low_fat).
```

```
diet(anemia, iron_rich).
```

```
diet(hypertension, low_salt).
```

```
suggest_diet(Disease, Diet) :-
```

diet(Disease, Diet).

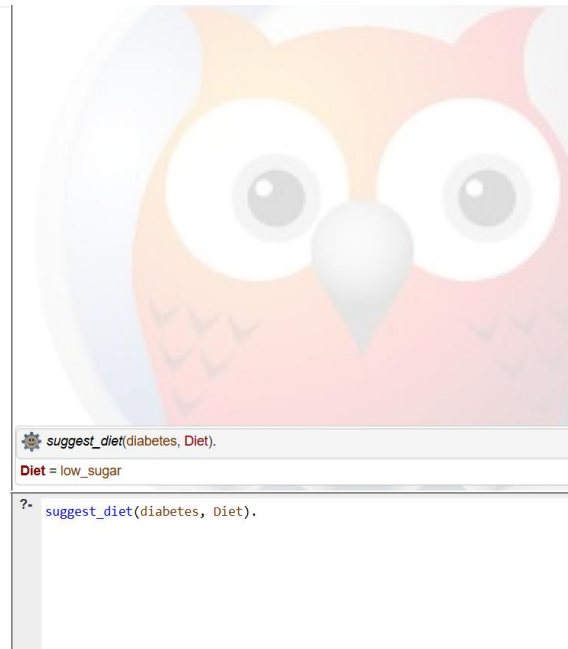
Query

suggest_diet(diabetes, Diet).

Output

Diet = low_sugar.

```
1 diet(diabetes, low_sugar).  
2 diet(obesity, low_fat).  
3 diet(anemia, iron_rich).  
4 diet(hypertension, low_salt).  
5  
6 suggest_diet(Disease, Diet) :-  
7   diet(Disease, Diet).
```



Result

Thus, the Prolog program to suggest a diet based on a given disease was successfully implemented.